EXCAVATIONS OVER 1.2 M IN DEPTH
TABLE 4-3
SIZE AND SPACING OF MEMBERS (S.I. [METRIC] UNITS)

UPRIGHTS			WALES		STRUTS			
					Min. Dimension (metres)		Maximum Spacing	
Trench Depth (metres)	Minimum Dimension (mm)	Maximum Spacing (metres)	Minimum Dimension (mm)	Maximum Vertical Spacing (metres)	Trench Width up to 1.8mm	Trench Width 1.8-3.7mm	Vertical (metres)	Horizontal (metres)
Hard and Solid Soils								
1.2-3	38×235	1.8	89×140	1.2	89×89	140×140	1.2	1.8
3-4.6	38×235	0.9	140×140	1.2	89×140	140×191	1.2	1.8
Soils Likely to Crack or Crumble								
1.2-3	38×235	0.9	89×140	1.2	89×140	140×140	1.2	1.8
3-4.6	38×235	0.9	140×191	1.2	140×140	140×191	1.2	1.8
Soft, Sandy, Filled or Loose Soils								
1.2-3	38×235	Close tight	140×191	1.2	140×140	140×191	1.2	1.8
3-4.6	38×235	Close tight	191×191	1.2	140×191	140×191	1.2	1.8

Notes on Table 4-2 apply to Table 4-3

Miscellaneous Hoisting Equipment

Miscellaneous Hoisting Equipment Requirements

4.18.1 Except as otherwise specified in the code, all cranes, derricks and similar hoisting equipment, shall be designed, constructed, erected, disassembled, maintained, and operated in accordance with the requirements of the manufacturer's specifications and instructions, and the following applicable standards, as amended from time to time

(1) Crane Manufacturers' Association of America, Inc. "Specifications for Electric Overhead Traveling Cranes", CMAA Specification #70, 1988,

(2) ANSI B-30.2-1990, "Overhead and Gantry Cranes,"

(3) ANSI B 30.11-1993, "Monorail Systems and Underhung Cranes,"

(4) CSA Standard Z248-1975 "*Code for Tower Cranes*,"

(5) CSA Standard Z150-1974 "*Safety Code for Mobile Cranes*,"
(6) ANSI ASME B30.7, 1994, "Base Mounted Drum Hoists,
(7) ANSI B30.16,-1993, "Overhead Hoists,"
(8) ANSI B30.6,-1990, "Derricks,"
(9) CSA Standard CAN/CSA-Z256-M87, "*Safety Code for Material Hoists.*"

Safe Working Load

4.18.2 Where the origin or the safe working load of a crane, derrick, or similar hoisting equipment cannot be ascertained, or its continued safe use cannot be assured, the equipment shall be re-certified by a professional engineer.

Professional Engineer's Certification

4.18.3 Any modification, manufacture, or repair of a structural element or component of a crane, derrick, or hoist shall be approved by the original manufacturer, or carried out under the direction of a professional engineer who shall certify the work performed.

Manufacturer's Manual

4.18.4 The manufacturer's manual for cranes, derricks, and similar hoisting equipment shall be available at the place where the equipment is used. The manual shall show the approved methods of erection, dismantling, operation, adjustment, and maintenance of the component parts and the assembled equipment.

Maintenance History

4.18.5 The manager shall maintain, or obtain a copy of, a file or record of the maintenance history of the following powered hoisting equipment used at the minesite

(1) overhead traveling bridge and gantry cranes,
(2) tower cranes,
(3) stiff leg, guy, A-frame and gin-pole derricks,
(4) material hoists, and
(5) mobile cranes, draglines, and other hoisting equipment of more than 1800 kg (4,000 lbs.) capacity.

Logbook

4.18.6 (1) A logbook shall be provided and maintained for each unit of equipment listed in section 4.18.5. All inspections, modifications, maintenance and repair work shall be recorded in the logbook, and dated and signed by the person who did the work.

(2) Any person involved in the operation of the hoisting equipment shall be responsible for recording in the logbook, any defect or operating

difficulty that the person notices, and reporting the matter to the person's supervisor.

(3) The supervisor shall examine, date and sign the logbook, at least weekly, to ensure the required entries have been made.

Name, Model and Serial Number

4.18.7 The name, model, and serial number assigned by the manufacturer shall be clearly visible and legible on every crane, hoist, and derrick.

Permanent Sign

4.18.8 (1) A permanent, clearly visible, and legible sign shall be affixed to each crane, derrick, and hoist showing the safe working load.

(2) The safe working load shall be marked on the load block and superstructure of hoisting equipment and shall be maintained in a legible condition.

(3) The safe working load shall not exceed the safe working capacity of any component in the hoisting system, including the supporting structure.

(4) Subsections (1) and (2) do not apply when the safe working load is affected by

(a) the length of a boom or jib, or its vertical or horizontal angle, or

(b) the position of a load-supporting trolley, or

(c) the use or position of outriggers to increase the stability of the unit.

(5) Where the safe working load is affected by the factors listed in subsection (4), a separate load chart showing the safe working loads in all possible working positions and configurations shall be permanently posted on the equipment or issued to the operator of the equipment. The operator shall keep the load chart in a legible condition and have it available at all times when operating the equipment.

Angle of Boom and Radius Lift

4.18.9 (1) Every crane, derrick, and similar hoisting equipment with a boom that can be moved in a vertical plane shall be provided with a device to indicate the angle of the boom or the radius of the lift.

(2) Telescoping booms shall be suitably marked to indicate the amount of boom extension or be provided with boom extension indicators.

(3) Devices installed in accordance with subsections (1) and (2) shall be clearly visible from the operator's location.

Controls

4.18.10 (1) Each control for a crane, derrick, or hoist shall be clearly marked to show its function.

(2) All bridge, gantry, and overhead traveling cranes operated by pendant or radio controls shall have signs on the crane structure, visible to an operator, clearly indicating the direction of hook, bridge, and trolley motions, compatible with those marked on the controls.

Radio Control Systems

4.18.11 Radio control systems shall meet the following requirements:

(1) they shall be designed to prevent movement of hoisting equipment except in response to actuation of the control devices, and only when all safety circuits are closed,

(2) provisions shall be made to maintain a signal-to- noise ratio of 10 to 1,

(3) multiple signals shall be generated by a minimum of two channels or by a coded single channel,

(4) where a code is used, two parts of the code shall be required for safety, neither of which can be easily generated from ambient signals, and a further information bit shall also be required to distinguish each signal,

(5) where two or more units may be operated in contact with each other, a further code signal shall be required to identify each unit and prevent response by other than the instructed unit, and

(6) the system manufacturer shall certify that the radio controlled system meets the above requirements.

Protection of Hoisting Equipment Operators

4.18.12 (1) Hoisting equipment operators shall be protected against heat, cold, flying, or falling objects or other adverse or dangerous conditions that could endanger them.

(2) Operator cabs shall afford protection from the weather and from overhead hazards. Windows shall be of safety glass or other material providing equivalent protection.

(3) A fire extinguisher, having a ULC rating of at least 5BC shall be immediately available to the operator of each cab-equipped crane.

Miscellaneous Hoisting Equipment – Operation

Crane Operators

4.19.01 (1) On or before June 1, 2018 operators of mobile cranes, boom trucks, folding boom and tower cranes performing critical lifts as defined in section 14.1 of the Occupational Health and Safety Regulation, B.C. Reg. 296/97, shall have a valid crane operator certification issued by a certifying agency acceptable to the chief inspector.

(2) On or before June 1, 2018, the manager shall ensure that workers performing occasional hoisting operations using mobile cranes, boom trucks, folding boom and tower cranes with a rated capacity equal to or greater than five tons or with a boom length of 25 feet or greater at a mine site are certified to Level D or equivalent of the British Columbia Crane Safety Association.

(3) The certification documents referred to in subsections (1) and (2) must be readily available for review by an inspector.

Operator's Inspection

4.19.1 The operator shall inspect the hoisting equipment at the beginning of the operator's shift and test the limit switches, brakes, circuit breakers, and other control and safety devices. If any defects or problems are noted, the operator shall notify the operator's supervisor and the equipment shall not be operated until the defects or problems have been corrected and authorization received from the supervisor.

Audible Warning Signal

4.19.2 An effective audible warning signal shall be provided on any hoisting equipment and operators shall ensure the signals are used as required.

Boom Stops

4.19.3 Where the design of a crane, derrick, or similar hoisting equipment is such that the boom could fall over backwards, positive boom stops shall be installed. In addition, a device shall be provided to automatically stop the hoisting of a boom when it reaches a predetermined angle.

Operator's Requirements

4.19.4 The operator of hoisting equipment shall

(1) not attempt to hoist any load if the operator has any doubt that it can be safely handled, but the operator shall inform the operator's supervisor who will determine what action is required,

(2) ensure a load being handled does not contact the boom and that the boom does not contact any structure or equipment,

(3) ensure the load block is suspended directly over any load being handled,

(4) ensure the loads shall be safely landed, supported, and stabilized before being unhooked, and

(5) not leave the controls of any hoisting equipment while a load is suspended.

Traveling With a Suspended Load

4.19.5 (1) When a crane is traveling with a suspended load, the operator shall ensure that the load is carried as close to the ground as possible and rigged, as necessary, to control swinging of the load.

(2) When necessary, a person designated as a signaler shall walk ahead of the moving load to warn persons to keep clear.

Authorized Person

4.19.6 (1) All loads shall be slung or hooked by an authorized person.

(2) Whenever the operator does not have an unobstructed view of the load hook and load throughout the whole range of the hoisting operation, the operator shall act only on the directions of an authorized signaler.

(3) Effective communication systems shall be used when weather conditions or distance render the use of hand signals impracticable or unsafe.

Contact With an Energized Electrical Conductor

4.19.7 Hoisting equipment, which has been in contact with an energized electrical conductor or struck by lightning shall be removed from service and not returned until safe use has been assured by a professional engineer.

Distance From High Voltage Conductors

4.19.8 No work shall be performed nor shall machinery, equipment, tools, or materials be used or stored within the following specified minimum distances from any high voltage electrical conductor or equipment capable of energizing the material or equipment.

Voltage (phase to phase)	Min. Distance
751 v to 75 kv	3 m (10 ft)
Over 75 kv to 250 kv	4.6 m (15 ft)
Over 250 kv to 550 kv	6 m (20 ft)

Two or More Cranes Used on a Lift

4.19.9 Where the use of two or more cranes, derricks and similar hoisting equipment is required in a lifting operation

(1) the operation shall be under the direct supervision of an authorized person, be fully planned in advance and the plans communicated to all persons involved in the operation, and

(2) effective communications shall be established and maintained between all persons involved, during the complete lifting operation.

Loads Not To Be Passed Over Persons

4.19.10 Loads shall not be passed over persons unless there is no practicable alternative, and then only after the persons have been warned of the danger.

Rigging

Use of Riggings and Fittings

4.20.1 Riggings and fittings shall be used only for the purposes for which they were designed and manufactured.

Maximum Rated Loads

4.20.2 Except as otherwise specified by this code

(1) the maximum rated load of any rigging or rigging assembly shall be warranted by the manufacturer of the equipment or by a professional engineer, and

(2) the maximum rated loads shall not exceed

(a) one-fifth of the ultimate breaking strength of the weakest component of the rigging, or

(b) one-tenth of the ultimate breaking strength of the weakest component of the rigging, when the rigging is used as a means of supporting workers.

Wedge Socket Connectors

4.20.3 When a wedge socket connector is used as a wire rope terminal, the end of the rope shall be secured to prevent release of the wedge, or rope slippage at the socket.

Open Hooks Not To Be Used

4.20.4 Open hooks shall not be used where dislodgement of the load from the hook would endanger persons.

Securement Against Dislodgement

4.20.5 Shackle pins, heel pins, and similar devices shall be secured against dislodgement.

Sheaves for Running Lines

4.20.6 Sheaves for running lines shall have a means of retaining the rope in the sheave groove.

Protection of Rigging Lines

4.20.7 Rigging lines, slings, and other components shall be protected against cutting, chaffing, and abrasion.

U-bolt Type Cable Clips

4.20.8 When U-bolt type cable clips are used for fastening wire ropes, the U-bolt shall be installed so that it bears on the short or "dead" end of the rope and in accordance with the manufacturers specifications.

Rope to Be Securely Fastened to the Drum

4.20.9 Unless a rope is required to automatically disengage from a winding drum, it shall be securely fastened to the drum and not less than three full turns of rope shall remain on the drum at all times. A rope shall not be secured to the inside of a drum by knotting or by affixing cable clips.

Removal of Hook From Service

4.20.10 A hook shall be removed from service when

(1) the original throat opening has increased by more than 15% as measured at the narrowest point, or

(2) it has twisted more than 10 degrees from its original plane, or

(3) it has cracks or other defects.

Rigging Removal From Service

4.20.11 Rigging shall be removed from service if it has contacted an electric arc, molten metal or other source of excessive temperature.

Safe Working Load

4.20.12 The safe working load of spreader bars and specialized lifting devices shall be certified by a professional engineer and clearly marked on the equipment.

Slings

Requirements

4.21.1 Except for slings which are field assembled in construction operations and which do not have swaged or pressed fittings, all slings made of chain, fiber webbing, wire rope, or which use swaged or pressed fittings, shall at the time of their assembly be permanently and legibly marked with the safe working load and the manufacturer's identification.

Lang's Lay Wire Rope Not to Be Used

4.21.2 Lang's Lay wire ropes shall not be used for slings.

Chain and Chain Slings

4.21.3 Chain and chain slings which have stretched or have links that are deformed, cracked, nicked, gouged, corroded, pitted, or burnt, shall be removed from service.

Fiber Webbing Slings

4.21.4 Fiber webbing slings with nicks, cuts, burns, or other damage or defect shall be removed from service.

Removal of Wire Ropes From Service

4.21.5 Wire ropes shall be removed from service when

(1) in running ropes, 6 randomly distributed wires are broken in 1 rope lay, or 3 wires are broken in 1 strand in any 1 lay, or

(2) in standing ropes and slings, there are more than 2 broken wires in 1 lay in sections between the end fittings.

TABLE OF CONTENTS

5.1	Electrical – General	3
5.2	Underground Mines	6
5.3	Cables	7
5.4	Switching Sequences	8
5.5	Control Systems	8
5.6	Underground Coal Mines and Hazardous Locations	8

Electrical – General

Codes and Standards

5.1.1 The manager must ensure that, unless modified by this code, all electrical equipment is installed, maintained and operated in accordance with CSA Standard M421 Use of Electricity in Mines, in conjunction with the *Canadian Electrical Code*, as amended from time to time.

Electrical Work

5.1.2 (1) The manager must ensure that design, installation, alteration and maintenance work performed under section 5.1.1 is only performed by, or under the supervision of, a certified person.

(2) The chief inspector may, for the purposes of subsection (1), require the certified person to be a professional engineer.

Electrical Safety Plan

5.1.3 (1) The manager must ensure that an electrical safety plan for the mine is developed, implemented and updated whenever relevant conditions or regulations change.

(2) The electrical safety plan must include training for workers who may be exposed to an electrical hazard while doing any of the following:

(a) performing maintenance on specialized equipment in the workplace, including

(i) electrolytic cells,

(ii) batteries, battery rooms or battery enclosures,

(iii) lasers,

(iv) power electronics equipment, or

(v) mobile electrical equipment;

(b) operating induced polarization geophysical systems as referred to in section 9.3.5 of this code.

(3) The manager must ensure that the electrical safety plan is kept at the mine and made available to an inspector on request.

Worker's Awareness

5.1.4 The manager must ensure that any person working near high voltage electrical equipment or conductors, whether exposed or potentially exposed during their work, is properly trained and familiar with

(a) the existence, location and voltage of the electrical equipment or conductors, and

(b) the work arrangements and procedures to be followed in accordance with the electrical safety plan referred to in section 5.1.3.

Notices and Information

5.1.5 (1) For any permanent electrical installation at a new mine, the manager must ensure that

(a) an electrical plan, approved by an electrical engineer registered within British Columbia, is developed for the use of electrical energy at the mine, and

(b) the electrical plan is forwarded to the chief inspector prior to the introduction of electricity at the mine.

(2) For any existing mine, the manager must ensure that

(a) a notification is sent to the chief inspector prior to the construction, expansion, alteration or replacement of fixed substations at the mine, and

(b) electrical plans respecting the actions referred to in paragraph (a) are kept at the mine and made available to an inspector on request.

Warning Notices

5.1.6 The manager must ensure that all vaults, rooms, enclosures or areas containing electrical equipment have a permanent, legible warning notice affixed in a conspicuous position at all points of entry or access which identifies the danger and indicates the highest voltage in use.

System Testing

5.1.7 (1) The manager must ensure that supply systems for mobile electrical equipment are tested before being put into service, and at least once a year thereafter, to prove the effectiveness of the ground fault tripping and the ground conductor monitoring circuits.

(2) The manager must ensure that a record of the tests required under subsection (1) are kept at the mine and made available to an inspector on request.

Electrical Test Equipment

5.1.8 The manager must ensure that electrical test equipment is

(a) inspected before each use, and

(b) maintained for safe operation and calibrated to ensure accuracy.

Electrically Insulated Elevating Work Platform

5.1.9 (1) The manager must ensure that any insulated elevating work platform that workers use to perform electrical related tasks is dielectrically tested in accordance with section 5.3.4 of Standard CAN/CSA-C225:20 “Vehicle Mounted Aerial Devices” at least once every 12 months.

(2) The manager must ensure that the insulating capability of the electrically insulated elevating work platform is certified by a CSA recognized testing agency.

(3) If the electrically insulated elevating work platform fails dielectric testing as set out in subsection (1), or if it is to be used only as a non-electrically insulated elevating work platform, the failure or use must be clearly identified on the platform, and on all of its related labeling, and in its records and logbook.

Submersible Pumps

5.1.10 The manager must ensure that a submersible pump that is supplied by a portable trailing cable is provided with ground fault protection to

(a) limit ground fault current to a maximum of 25 amps, and

(b) automatically isolate the supply in the event of a ground fault.

Low Voltage Disconnection and Lockout

5.1.11 (1) The manager must ensure that low voltage electrical equipment is completely disconnected and locked out before starting work on it.

(2) Despite subsection (1), if adjustments, tests, calibrations, repairs or alterations with respect to electrical conductors or circuit parts cannot be performed in a de-energized state because of equipment design or operational limitations, or de-energizing will introduce new hazards or increase existing hazards,

(a) energized work may be performed, and

(b) all necessary precautions consistent with the nature and extent of the hazards must be taken to ensure that the work can be performed safely.

Electrical Rooms

5.1.12 (1) The manager must ensure that electrical rooms are located and constructed to have the best practical protection against fire propagation, water and dust ingression and corrosive atmospheres.

(2) The manager must ensure that access to electrical rooms is restricted to authorized personnel only.

(3) The manager must ensure that electrical rooms are sufficiently ventilated to maintain the electrical equipment inside at safe temperatures.

(4) The manager must ensure that clear access to electrical equipment is maintained at all times.

(5) The manager must ensure that electrical rooms

(a) are not used for general storage,

(b) store only equipment and materials directly related to the electrical system,

(c) do not store flammable materials, including combustible liquids and gases, and
(d) are not used as a workstation for work not related to the electrical equipment in the room.

(6) The manager must ensure that items stored in an electrical room do not obstruct egress routes or interfere with ventilation systems, fire protection systems or any other safety-critical infrastructure within the electrical room.

(7) The manager must ensure that items such as spare fuses, arc flash personal protective equipment, high voltage tools and draw-out type breakers are stored in appropriate storage areas.

(8) The manager must ensure that each electrical room has a single-line diagram posted showing, for that electrical room,
(a) the relative locations of all fixed electrical distribution equipment,
(b) the ratings of all fixed electrical distribution equipment, and
(c) the routes, properly noted and referenced, of all fixed power distribution feeders.

(9) The manager must ensure that the single-line diagrams referred to in subsection (8) are reviewed and updated at least annually and after any significant change to the electrical system.

Isolating Surface Workings

5.1.13 The manager must ensure that the operation of high voltage switchgear is conducted by a qualified person.

Underground Mines

Isolating Underground Workings

5.2.01 (1) The manager must ensure that switchgear is
(a) installed at a convenient location on surface to provide the means for isolating all underground circuits, and
(b) operated by a qualified person.

(2) The manager must ensure that switchgear installed underground is
(a) built of non-combustible materials, and
(b) fixed in a vertical position on a metal frame or plate.

(3) The manager must ensure that switchboards located underground are
(a) recessed from hallways, and
(b) have a floor on which water cannot accumulate.

5.2.1 Repealed.

5.2.2 Repealed.

5.2.3 [Renumbered as 5.6.3]

5.2.4 [Renumbered as 5.6.4]

Cables

Handling Trailing Cables

5.3.1 [Renumbered as 5.6.5]

5.3.2 [Renumbered as 5.6.6]

5.3.3 [Renumbered as 5.6.7]

5.3.4 In accordance with sections 1.8.1 and 1.8.2,

(a) the manager must ensure that appropriate protective equipment is provided to a person handling energized trailing cables at the mine, and

(b) the person must use the equipment as instructed by a qualified person and ensure it is in good working order.

Defective Trailing Cables and Associated Connection Accessories

5.3.5 The manager must ensure that defective trailing cables and associated connection accessories are

(a) inspected by a certified person to confirm they are in safe operating condition before they are returned to service, or

(b) if paragraph (a) does not apply, identified and removed from service.

Damaged Trailing Cables and Associated Connection Accessories

5.3.6 (1) The manager must ensure that damaged trailing cables and associated connection accessories are, before being returned to service,

(a) repaired by a certified person or by a qualified person under the supervision of a certified person, and

(b) tested by a certified person.

(2) The manager must ensure that a record of the tests required under subsection (1) (b) are kept at the mine and made available to an inspector on request.

Armored Cables

5.3.7 The manager must ensure that at the mine site,

(a) armored cables are not used in place of portable power cables or flexible cords, and

(b) armored cable connections or terminations are not made using any type of quick connect couplers.

Single-conductor Cable Connections

5.3.8 (1) The manager must ensure that single-conductor quick connect couplers are not used on three-phase electrical systems at the mine site.

(2) Despite subsection (1), a single-conductor quick connect coupler may be used on a three-phase electrical system at the mine site if the

coupler is protected from being disconnected while energized through one of the following means:

(a) the coupler is mechanically or electrically interlocked with the upstream disconnect switch in a way that ensures the coupler can only be engaged or disengaged while de-energized;

(b) through the use of a trapped key lock method, or a substantially similar method, the coupler is rendered inaccessible to persons while it is energized.

Switching Sequences

Switching Plan

5.4.01 (1) The manager must ensure that a written switching plan is developed, reviewed and implemented prior to the coordinated operation of high voltage electrical switchgear.

(2) The written switching plan must include an up-to-date and detailed electrical single-line diagram, the sequence of operations and the devices to be operated and any corresponding steps.

5.4.1 Repealed.

Control Systems

General Requirements for Control Systems

5.5.01 The manager must ensure that

(a) the risk to workers posed by electrical equipment with control systems is minimized,

(b) the controlled equipment cannot be inadvertently activated,

(c) emergency stop devices are hardwired, and

(d) after a power interruption or low voltage occurrence, automatic startup is prevented if automatic startup in such circumstances is likely to create a hazard for workers.

5.5.1 Repealed.

5.5.2 Repealed.

5.5.3 Repealed.

Underground Coal Mines and Hazardous Locations

Underground Coal Mines and Hazardous Locations

5.6.1 Repealed.

5.6.2 Electrical energy used in an underground coal mine or in a hazardous location must conform to the requirements in the *Canadian Electrical Code*.

Underground Coal Equipment Certification

5.6.3 (1) The manager of a coal mine, when installing electrical equipment underground, shall

(a) if the equipment is new, ensure that it has been certified as suitable for use in an underground coal mine by an accredited laboratory and complies with the relevant standards as produced by one of the following organizations and as accepted by the chief inspector:

(i) Canadian Standards Association (CSA);
(ii) Natural Resources Canada (NRCan);
(iii) United States Mine Safety and Health Administration (MSHA);
(iv) International Electrotechnical Commission (IEC);
(v) International Organization for Standardization (ISO);
(vi) European Committee for Electrotechnical Standardization (CENELEC).

(b) If the equipment is not new and has been obtained from a source outside of the manager's control, ensure that it has been certified by an accredited laboratory and has been assessed and complies with one of the standards listed in paragraph (a) in its current configuration.

Hazardous Locations

5.6.4 The manager shall ensure that all electrical equipment used in a hazardous location, as defined in the *Canadian Electrical Code*, is approved for use in such a location and for the specific gas, vapour, or dust that is or may be present.

Manager's Responsibility

5.6.5 The manager of a coal mine shall

(1) develop an inspection and maintenance schedule for all electrical equipment in use underground or in any hazardous location, as defined by the *Canadian Electrical Code*, and

(2) designate qualified persons to make the inspections and carry out the maintenance as described in the approved schedule.

Repair of Power System

5.6.6 At any place in an underground coal mine, or in any hazardous location where flammable gas could accumulate, the repair, adjustment, or replacement of electrical equipment shall only be carried out

(1) after the equipment has been disconnected from the power supply and is electrically dead, and

(2) in a location where the electrician doing the work is satisfied that no dangerous concentration of flammable gas is present.

Flammable Gas Warning

5.6.7 In any location where flammable gas could accumulate in dangerous amounts, the manager shall ensure that an approved automatic gas detector is available to continually monitor the air at that location. The monitor shall be of a type that will give an audible or visual warning whenever a predetermined percentage of flammable gas is present.

5.7.1 Repealed.

5.7.2 Repealed.

TABLE OF CONTENTS

6.1	Work System Approval	3
6.2	Notification	3
6.3	Manager – Underground	3
6.4	Mine Shiftboss – General Rules	5
6.5	Shiftboss – Surface Mine	5
6.6	Shiftboss – Underground	6
6.7	Fireboss – Underground Coal	6
6.8	Mine Plans	6
6.9	Mine Haul Road Design	7
6.10	Dumps	8
6.11	Underground Mine Design Features	9
6.12	Mine Openings	9
6.13	Refuge Stations	10
6.14	Repealed	10
6.15	Track Haulage Systems	11
6.16	Underground Mobile Equipment	11
6.17	Safety Stations	11
6.18	Remote Control Operations and Emerging Technology	11
6.19	Equipment Operation – General	14
6.20	Haulage Vehicle Operations – Surface	15
6.21	Battery Charging	17
6.22	Crew Transportation – Rail Track	17
6.23	Surface Mines	17
6.24	General Mine Rules	18
6.25	General – Underground Rules	18
6.26	Rock Passes	19
6.27	Stairways	20
6.28	Ladders	20
6.29	Raise – Underground	21
6.30	Flammable Waste & Scrap – Underground	22

6.31	Sulphide Dust	22
6.32	Fire Doors – Underground	22
6.33	Ventilation Standard	23
6.34	Main Ventilation – Interruptions	23
6.35	Underground Mine Heating	23
6.36	Auxiliary Ventilation	23
6.37	Ventilation Monitoring	24
6.38	Doors and Regulators	25
6.39	Quantities – Splits – Measurement for Coal Mines	25
6.40	Unventilated Workings	25
6.41	Stoppings – Coal Mines	26
6.42	Flammable Gas	26
6.43	Prevention of Ignition	27
6.44	Combustible Dust – Coal Mines	28
6.45	Explosion Barriers – Coal Mines	31
6.46	Explosion Barriers – Other Mines	31

Work System Approval

Qualified Persons

6.1.1 (1) The mine manager shall ensure the design of the mine excavations, openings, support systems, fixtures, features, methods of operation and all other works necessary to operate a mine meet acceptable standards of practice and are carried out under the authority of a qualified person or persons.

(2) At every underground mine the manager shall appoint a qualified person to be responsible for all aspects of ventilation in the mine.

(3) The chief inspector may for the purposes of 6.1.1 require the qualified person to be a professional engineer or other licensed professional as may be appropriate.

Notification

Notice To Start Work

6.2.1 The manager shall give 10 days' notice to an inspector of intention to start work in, at, or about a mine, including seasonal reactivation.

Notice to Stop Work

6.2.2 The manager shall give notice to an inspector of intention to stop work in, at, or about a mine, permanently, indefinitely, or for a definite period exceeding 30 days, and except in an emergency, the notice shall be not less than seven days.

Manager – Underground

Duty to Post Plans

6.3.1 (1) The manager of every underground mine shall post, or have posted, in a conspicuous place accessible to all persons working underground, a copy of the current emergency and rescue plan required by section 6.3.1(2), and a copy shall be sent to the OHSC.

(2) An emergency and rescue plan of the workings of the mine, satisfactory to the manager and which is in general conformance to accepted standards of practice, on which the main routes of egress from the active workings, sites of all underground refuge stations, and ventilation as required in section 6.3.2(5).

Copy for Inspection

6.3.2 The manager of an operating underground mine shall, upon request,

(1) supply an inspector with an accurate and up-to-date copy of the plan of the underground workings,

(2) a copy of the current emergency and rescue plan required by 6.3.1(2),

(3) a separate underground plan of every level showing all workings, including shafts, tunnels, diamond drill holes, dams, bulkheads, electrical substations, explosives storage, shop areas, permanent seals and stoppings,

(4) vertical mine sections at suitable intervals and azimuths, showing all shafts, tunnels, drifts, sloped roadways, rooms, stopes, diamond drill holes and the location of the top of the bedrock, surface of, and type of, the overburden and the bottom and surface of any known body of water or watercourse or material likely to flow,

(5) ventilation plan showing the normal direction and volume of the main air currents and the location of permanent fans, ventilation doors, fire doors, stoppings and connections with adjacent mines, and

(6) a plan indicating the position of all fixed electrical apparatus in the mine and the routes of all fixed power feeders and fixed branch feeders, properly rated and referenced, and the rating of all electrical feeder control apparatus and equipment.

Shift Boss Responsibility for Plans

6.3.3 The manager shall ensure each shiftboss is provided with current development plans for the assigned area of responsibility indicating the size, dip and length of all development openings.

The plans shall:

(1) Clearly indicate points where openings, adjacent, old or disused workings are or will be within 30m and 8 m of a breakthrough for the purposes of section 8.10.2.

(2) Clearly indicate points where diamond drill holes may be within 8 meters of a mine opening.

(3) Exploration drill holes that extend into mine travelways and work places from areas where blasting operations may take place shall be marked and effectively blocked.

(4) Exploration drill holes shall have their collars located and the holes shall be plugged as required.

6.3.4 The shiftboss shall ensure the mining crews are informed of those possible breakthrough points in accordance with section 6.3.3.

6.3.5 The manager shall ensure that effective ground support systems are properly installed and maintained, and that a quality control program is in place to ensure that the ground support systems are installed and maintained according to the directions of a qualified person.

Mine Shiftboss – General Rules

Examination of Workings

6.4.1 (1) All active workings shall be examined by the certified shiftboss or supervisor with assigned responsibility to ascertain that they are in a safe working condition, as often as the nature of the work necessitates.

(2) All persons working underground shall have their work areas inspected by a shiftboss or supervisor at least twice per shift.

(3) Subject to subsection (4), work areas containing underground autonomous or semi-autonomous tracked or rubber-tired mobile equipment must be inspected by the shiftboss or supervisor prior to employees entering the work area, or at least once per shift.

(4) In an underground coal mine, the fireboss must conduct the workplace inspections required by subsection (3).

Daily Examination and Report Book

6.4.2 The person making the examination under section 6.4.1 shall before going off shift record all unusual and hazardous conditions and corrective actions taken or proposed in a daily examination and report book, and sign the report as a record of the conditions found. For underground mines the record shall include a report on each working place examined.

6.4.3 The report made under section 6.4.2 shall be read and countersigned by the corresponding supervisor on the oncoming shift and the unusual and/or hazardous conditions discussed with the workers before they are permitted to resume operations in the areas indicated in the record.

Shiftboss – Surface Mine

Pit Face Area Safety

6.5.1 No work shall be carried on, at, or below a face or wall of a surface mine until that face or wall has been examined and declared safe by the shiftboss or, in the case of a mine with fewer than 6 employees, the supervisor.

Dump Block

6.5.2 A dump or stockpile area on the surface of a mine shall be examined by either the open pit shiftboss or a qualified person

(1) before material is permitted to be dumped where dumping has not been carried on for a period of 4 hours or more, and

(2) at least once and at intervals of not more than 4 hours during each shift when material is being dumped, and